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Key Points:

- High-resolution differentiable models offer substantial and widespread improvements for CONUS streamflow prediction accuracy
- With a differentiable routing model, we produced a seamlessly discretized streamflow product across approximately 180,000 CONUS river reaches
- Multiscale training can resolve spatial heterogeneity and improve peak flow, while explicit routing enables large-river simulations

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High-Resolution National-Scale Water Modeling Is Enhanced by Multiscale Differentiable Physics-Informed Machine Learning

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Abstract The National Water Model (NWM) is a key tool for flood forecasting, planning, and water management. Key challenges facing the NWM include calibration and parameter regionalization when confronted with big data. We present two novel versions of high-resolution (~37 km²) differentiable models (a type of hybrid model): one with implicit, unit-hydrograph-style routing and another with explicit Muskingum-Cunge routing in the river network. The former predicts streamflow at basin outlets whereas the latter presents a discretized product that seamlessly covers rivers in the conterminous United States (CONUS). Both versions use neural networks to provide a multiscale parameterization and process-based equations to provide a structural backbone, which were trained simultaneously (“end-to-end”) on 2,807 basins across the CONUS and evaluated on 4,997 basins. Both versions show great potential to elevate future NWM performance for extensively calibrated as well as ungauged sites: the median daily Nash-Sutcliffe efficiency of all 4,997 basins is improved to around 0.68 from 0.48 of NWM3.0. As they resolve spatial heterogeneity, both versions greatly improved simulations in the western CONUS and also in the Prairie Pothole Region, a long-standing modeling challenge. The Muskingum-Cunge version further improved performance for basins >10,000 km². Overall, our results show how neural-network-based parameterizations can improve NWM performance for providing operational flood predictions while maintaining interpretability and multivariate outputs. The modeling system supports the Basic Model Interface (BMI), which allows seamless integration with the next-generation NWM. We also provide a CONUS-scale hydrologic data set for further evaluation and use.

1. Introduction

Floods have caused more damage and financial losses than other natural hazards, amounting to an estimated US \$32.1 billion across the country in 2020 (Wing et al., 2022), and the impacts are continuing to grow. The National Water Model (NWM) aims to provide actionable information for flood and drought forecasting and water resource management to protect lives and property and enhance the national economy (Cosgrove et al., 2024). Furthermore, headwater and downstream communities exhibit different sensitivities to climate risks and may respond differently to extreme events due to scale-dependent water availability and environmental covariations. Consequently, large-scale assessments and management strategies may overlook the unique vulnerabilities of local communities (Dooze, 1986; Ke & Zhang, 2024; Song, Shen, et al., 2024).

The current NWM has a versatile modeling framework consisting of an integrated hydrologic model (WRF-Hydro), a meteorological pre-processor (called the Meteorological Forcing Engine), a geospatial hydrofabric (a set of scaled geospatial model inputs representing the geometry and topology of river networks, the land surface, and the shallow subsurface), and parameter calibration and regionalization tools (Cosgrove et al., 2024). WRF-Hydro has multiple physical process modules, including a column land surface model updated from the community NoahMP land surface model with 1 km resolution (Niu et al., 2011; Yang et al., 2011), overland and

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shallow saturated subsurface flow routines at 250 m resolution (Julien et al., 1995), a conceptual baseflow bucket model at the unit level of NHDPlus catchments, a channel flow routing module using the Muskingum-Cunge method (Read et al., 2023), and a simple fill-and-spill reservoir module. The first version of the NWM (v1.0) has provided useful outputs for the conterminous United States (CONUS) since 2016. Updated versions have incorporated parameter calibration (v1.1), streamflow data assimilation (DA) (v1.2), reservoir representation (v2.0) (Johnson et al., 2023), and a total water level coastal coupling component (v3.0). Spatially, it has also been expanded beyond the CONUS to include Hawaii (v2.0), Puerto Rico (v2.1), and Alaska (v3.0) (Cosgrove et al., 2024). The NWM features various operational configurations, such as Analysis and Assimilation (open-loop retrospective simulations without DA and simulations with streamflow DA) as well as CONUS Short-Range (18h), Medium-Range (10d), and Long-Range (30d) ensemble forecasts.

Despite their explanatory power, traditional physical models, including the NWM, continue to be constrained by parameter calibration and structural updating strategies that cannot effectively absorb information from large data. This limitation is particularly evident from the challenge in addressing issues related to scaling and the difficulty in obtaining generalizable models (Peters-Lidard et al., 2017), which can manifest in limited performance improvements even when more data become available. The perpetual challenge of parameter calibration and regionalization is more evident when we want to incorporate information from thousands of gages across the CONUS. Parameter calibration is performed basin by basin over a certain time span on selected data-rich basins. Then the calibrated parameters are directly transferred to data-sparse regions after regionalization (Cosgrove et al., 2024), which can induce significant uncertainty in predictions at ungauged locations (Fang et al., 2024). This calibration strategy is time-consuming and prone to both local overfitting and poor spatial generalization.

Recently, differentiable models (DMs) (Feng et al., 2022; Shen et al., 2023) have made substantial progress as powerful simulation and big-data learning tools in hydrology and earth system modeling due to their high accuracy, great efficiency in learning from big data, full interpretability, and strong extrapolation capabilities. Unlike physics-informed neural networks (PINNs) (Raissi et al., 2019; Tartakovsky et al., 2020)—purely neural network-based models acting as surrogates for physical models with physical constraints incorporated into the loss functions—DMs are real physical models that integrate neural networks (NNs) to infer parameters or replace specific components. Although using an NN as a surrogate model (Tsai et al., 2021; Zhong et al., 2024) can somewhat link NNs and process-based equations, having the process equations explicitly implemented provides significant benefits, including enabling further updates to model structures and learning challenging relationships like scaling functions (Song, Shen, et al., 2024). In DMs, the physical model's equations are explicitly implemented to serve as a structural backbone and the NNs support it by providing physically-meaningful parameters or missing processes. The entire system is differentiable and trained as a unified pipeline, enabling end-to-end optimization via backpropagation. Differentiability, often supported by automatic differentiation inherent to machine learning platforms such as Pytorch, TensorFlow, and JAX, enables easy and efficient gradient tracking across many basins simultaneously. Additionally, user-defined adjoint functions can enable the differentiability of implicit numerical solvers (Bandai, 2022; Jay-Allemand et al., 2020; Rackauckas et al., 2021; Song, Knoben, et al., 2024). DMs not only show a high level of accuracy compared to state-of-the-art pure machine learning models but also maintain the full interpretability of the physical processes due to their seamless integration of the physical models (Aboelyazeed et al., 2023; Feng et al., 2022; Shen et al., 2023; Song, Knoben, et al., 2024). Beyond predicting the variables on which they are trained, DMs also provide good performance for untrained output variables because physical processes provide the linkages. These physical constraints also lead to better extrapolation abilities than pure ML models, enhancing predictions for ungauged basins (Feng et al., 2023; Hrachowitz et al., 2013) and unseen extreme events (Song, Sawadekar, et al., 2024). Another advantage of DMs is their capability for efficient whole-domain parameter regionalization facilitated by NNs, highly parallel computations, and training using gradient descent. They can rapidly assimilate big data (thus imposing strong constraints that work well on many sites) to produce robust, generalizable, and highly-performant parameterization schemes (Feng et al., 2023; Jiang et al., 2020; Li et al., 2024; Song, Sawadekar, et al., 2024; Wang et al., 2024; Zhong et al., 2024).

While DM approaches have shown promise to offer strong operational solutions towards boosting national-scale flood predictions, several obstacles remained. In particular, previous DMs mostly were “coarsely spatially lumped,” requiring the definition of the whole upstream catchment and basin-averaged inputs, and were thus not able to resolve the fine-scale heterogeneity in meteorological forcings and basin attributes. They could only output lumped basin-averaged fluxes, states, or streamflow at the outlet of the basin (Feng et al., 2022; Song, Sawadekar,

et al., 2024), similar to the lumped modeling approach used operationally at River Forecast Centers (R. A. Clark et al., 2014). Oftentimes spatially-distributed outputs over all river reaches are operationally desired to predict inundation. Previously, the coarse DMs used a unit-hydrograph-style routing formula for basin-scale routing. An external routing model capable of routing high-resolution runoff across the entire continental river network is essential for generating a seamless high-resolution streamflow product. Wang et al. (2024) developed and performed a differentiable distributed hydrological model in the Amazon Basin with grid cells at a resolution of 0.5° on a daily scale, where the streamflow is routed across grid cells. A connected neural network was used to provide grid-specific parameters for the physical module. Similarly, Zhong et al. (2024) developed a differentiable distributed model based on the river network of the sub-basins in the Source Region of the Yellow River.

Here, learning from the NWM's high-resolution approach and leveraging some of its software infrastructure and data pipeline, we developed a multiscale parameterization scheme to produce seamlessly discretized simulations across the entire high-resolution CONUS river network. We incorporated a differentiable Muskingum-Cunge model for large-scale routing over the whole CONUS, expanding on our previous work (Bindas et al., 2024). We performed a comprehensive evaluation across the CONUS, comparing our model's streamflow, snow water equivalent (SWE), and peak flow predictions with those from past versions of the NWM to show the improvements offered by DM. Additionally, we have provided seamless products for streamflow, SWE, soil moisture, evapotranspiration, and base flow at the unit basin level, utilizing the Multi-Error-Removed-Improved-Terrain (MERIT) Hydro DEM based unit basins data set (Lin et al., 2021). Only streamflow and SWE products were validated here, and other variables are provided only for evaluation purposes. Due to space constraints and earlier comparisons to NNs, this study focuses mainly on evaluating the physical models (NWM and DMs) and not purely data-driven models. The DMs have demonstrated accuracy that is highly competitive with purely data-driven models while providing full physical interpretability. Some comparisons with long short-term memory (LSTM)-based models have already been provided elsewhere (Feng et al., 2023; Song, Sawadekar, et al., 2024) and a more comprehensive comparison between DMs and LSTM networks will be addressed in our forthcoming work. In this work, we focused on the progress of this particular model in comparison to previous versions of the NWM over thousands of sites across the CONUS, and sought answers to the following scientific questions:

1. Can an NN-supported parameterization with a high resolution and multiscale training scheme significantly improve operational streamflow prediction for thousands of basins across the CONUS?
2. Where does the newly introduced multiscale training scheme have advantages and disadvantages?
3. Does incorporating an external differentiable Muskingum-Cunge routing model further enhance the streamflow simulation performance of the high-resolution DM?
4. How can DMs contribute to and be integrated into the development of the next generation of the National Water Model?

2. Methods

2.1. Data

Multi-Error-Removed-Improved-Terrain (MERIT) Hydro is a high-resolution vector hydrography data set derived from elevation data, which describes the global river network and divides it into 2.97 million reaches, including 180 thousand reaches across the CONUS (Lin et al., 2021). When we refer to “MERIT unit basins,” we refer to those defined in the unit basin product derived from MERIT-Hydro, which have a minimum size of 25 km^2 and a median size of 37 km^2 across the CONUS. Meteorological forcings, including basin-averaged precipitation as well as maximum and minimum temperatures for each MERIT unit basin, were derived from Daymet version 4 R1 (Thornton et al., 2022). Potential evapotranspiration was estimated using the method from Hargreaves (1994) with daily temperature and latitude (detailed in Table B1 of Appendix B). The static attributes, including topography, climate patterns, land cover, soil composition, and geological characteristics, were derived from diverse sources which can be found in Table A1 and A2 in Appendix A.

For model evaluation, we selected 4,997 basins within the CONUS where National Water Model version 3.0 (NWM3.0) streamflow simulation data were available (NOAA, 2024) and where valid daily observation data availability from the Geospatial Attributes of Gages for Evaluating Streamflow, version II (GAGES-II) data set (Falcone, 2011) exceeded 10%. In addition to the streamflow evaluation, in-situ SWE observations from 525

Snowpack Telemetry (SNOTEL) sites (Serreze et al., 1999) were used for a point-wise assessment of an untrained model output variable, SWE.

2.2. DMs

We discuss three DMs in this work. The first model, referred to as δ HBV1.0-CONUS-hydroDL, is a coarse lumped model presented in Feng et al. (2022) with a unit hydrograph routing formula (Section 2.2.1). The second model, termed δ HBV2.0UH-CONUS-hydroDL, is a high-resolution MERIT-unit-basin-based differentiable hydrologic model using the unit hydrograph routing formula (Section 2.2.2). The third model (δ HBV2.0 δ MC-CONUS-hydroDL) is similar to the second one but employs an external differentiable Muskingum-Cunge method for routing across the continental river network (Section 2.2.3). Here we will focus on the two high-resolution models. The codes of δ HBV1.0 and δ HBV2.0 are implemented in a domain-agnostic, PyTorch-based framework, δ MG, for developing trainable DM (Lonzarich et al., 2025).

2.2.1. Coarse Basin-Lumped Differentiable Streamflow Model (δ HBV1.0)

The coarse differentiable hydrological model (hereafter referred to as δ HBV1.0) is built on the work of Feng et al. (2022), which was further improved by Song, Knoben et al. (2024): a neural network is utilized for regional parameterization and the conceptual model Hydrologiska Byråns Vattenbalansavdelning, called HBV (Agha-kouchak & Habib, 2010; Beck et al., 2020; Seibert, 1997), is employed for hydrologic modeling. The HBV model is a bucket-based hydrological model that incorporates processes such as snowmelt, snow accumulation, infiltration, and runoff generation. The details of the processes and parameters of the HBV model are provided in Appendix B and can also be found in Feng et al. (2022) and Song, Knoben et al. (2024). The structure of δ HBV1.0 (Figure 1a) can be summarized as:

$$\theta_{d,b}^{1:t}, \theta_{s,b} = \text{LSTM}_w(\mathbf{x}_b^{1:t}, \mathbf{A}_b) \quad (1)$$

$$\mathbf{q}'_b^{1:t} = \text{HBV}(\theta_{d,b}^{1:t}, \theta_{s,b}, \mathbf{x}_b^{1:t}) \quad (2)$$

$$\mathbf{q}_b^{1:t} = \text{UH}(\mathbf{q}'_b^{1:t}, \theta_{ra}, \theta_{rb}) \quad (3)$$

where $\theta_{d,b}$ and $\theta_{s,b}$ represent dynamic (subscript d) and static (subscript s) HBV parameters lumped on the entire gage basin (denoted by subscript b), respectively. $\theta_{d,b}$ includes the time-variant shape coefficients, of effective flow (β) and evapotranspiration (η), as used in Feng et al. (2022). The other parameters in HBV ($\theta_{s,b}$) and unit hydrograph (UH) routing (θ_{ra} and θ_{rb}) are static (listed in Table B1 of Appendix B). The unit hydrograph routing formulas are shown in Equations 8 and 9. Subscript w represents the weights and biases of LSTM. The structures of LSTM are provided in Appendix B. $\mathbf{A}_b$ contains the average static attributes of the basin, including topography, soil texture, land cover, and geology (listed in Table A1 of Appendix A). $\mathbf{x}_b$ represents the lumped meteorological forcings (precipitation, mean temperature, and potential evapotranspiration) of the basin. $\mathbf{q}'_b$ and $\mathbf{q}_b$ are the simulated streamflows before and after routing, respectively, with units of mm/day. $\mathbf{q}_b$ can be converted to $\mathbf{Q}_b$ with units of m^3/s using the basin area.

As described in Feng et al. (2022) and Song, Knoben et al. (2024), the neural network outputs 16 sets of HBV parameters as 16 components of HBV, resembling subbasin-scale “hydrologic response units” in many hydrologic models or, in a computational sense, the “hidden units” used in the LSTM network. Note that this differs from the random seed ensemble approach used for LSTM (Kratzert et al., 2021), as only one neural network and one base model structure is used; future work will incorporate more ensemble members for a performance boost. While the coarse model utilizes basin-lumped inputs extracted for the entire upstream basin of the gage (called “gage basins” here to differentiate them from the high-resolution unit basins in MERIT, as shown in Figure 1) to predict streamflow and has demonstrated similar performance to LSTM on the CAMELS data set, its coarse resolution overlooks the heterogeneity of inputs within large basins.

2.2.2. A Multiscale DM With Unit Hydrograph Routing (δ HBV2.0UH)

We developed and trained a new model with inputs and outputs defined at the scale of MERIT unit basins (median size 37 km^2 over the CONUS) rather than the coarser gage basins (size ranging from 12 to $50,000 \text{ km}^2$, with a

median of $\sim 700 \text{ km}^2$). The runoffs generated by MERIT unit basins are aggregated and then routed to the basin outlet by the unit hydrograph routing formulas, then compared to observations during training (Figure 1b). This model is called $\delta\text{HBV2.0UH-CONUS-hydroDL}$ ($\delta\text{HBV2.0UH}$ for short).

A multiscale parameterization is used for high-resolution MERIT unit basins in $\delta\text{HBV2.0UH}$, which is conceptually similar to the multiscale soil moisture model in J. Liu et al. (2022) and also influenced by the philosophy in the literature (Rakovec et al., 2016, 2019). However, this multiscale parameterization is supported by more powerful NNs and differentiable equation solvers. “Multiscale” means the parameters are estimated at the small MERIT unit basin level, while the training (more precisely, the calculation of the loss function) occurs at the larger gage basin scale. Some works (Huynh et al., 2024; Wang et al., 2024) have implemented a gridded version of a DM where routing is integrated during training. However, routing can be computationally expensive at high resolution and the national scale, which hinders learning from a large data set. Thus, our training happens in two separate stages: rainfall-runoff processes at a daily interval and river routing at an hourly interval, akin to the training method from Zhong et al. (2024). This model can capture heterogeneity within large basins through the use of high-resolution inputs and, importantly, can constrain the parameterization at a fine scale so model behavior is consistent between training and forward simulations at the MERIT unit basins. The regional parameterization formula of δHBV can be summarized as:

$$\theta_{d,m}^{1:t} = \text{LSTM}_w(\mathbf{x}_{b,m}^{1:t}, \mathbf{A}_m) \quad (4)$$

$$\theta_{s,m} = \text{MLP}_w(\mathbf{A}_m) \quad (5)$$

where $\theta_{d,m}^t$ and $\theta_{s,m}$ are dynamic (subscript d) and static (subscript s) HBV parameters, respectively, for a MERIT unit basin (denoted by subscript m). $\theta_{d,m}^t$ (specific to time t) includes the shape coefficients, of effective flow (β) and evapotranspiration (η) and a dynamic recession coefficient (k_0) of fast flow, as used in Song, Sawadekar et al. (2024). The other HBV parameters and the unit hydrograph routing parameters used in Equations 8 and 9 are static and included in $\theta_{s,m}$. $\mathbf{A}_m$ contains the static attributes of the MERIT unit basins, including topography, soil texture, land cover, and geology attributes, as listed in Table A1 in Appendix A. $\mathbf{x}_m$ represents the meteorological forcings for the MERIT unit basin, including precipitation, mean temperature, and potential evapotranspiration. Unlike the previous versions that use an LSTM network for both dynamic and static parameters, we employed an LSTM (LSTM_w) and a multilayer perceptron (MLP_w) network separately to distinguish the parameterizations for dynamic and static parameters. Subscript w represents their weights. MLP_w was used for static parameters to reduce the GPU memory usage required by LSTM_w but cannot output dynamic parameters. The structure of the LSTM is the same as that used in the lumped model, $\delta\text{HBV1.0}$, and is described in Appendix B along with the structure of the MLP. Only 4 components of HBV with 4 sets of HBV parameters from the NNs were used in the high-resolution model ($\delta\text{HBV2.0}$), to reduce the GPU memory usage. One should note that all inputs and parameters are at the MERIT-unit-basin resolution.

The hydrological processes on the MERIT unit basins simulated by the HBV model can be summarized as:

$$\mathbf{q}_m^{1:t} = \text{HBV}(\theta_{d,m}^{1:t}, \theta_{s,m}, \mathbf{x}_m^{1:t}) \quad (6)$$

$$\mathbf{q}_b^{t'} = \sum_{i=1}^n \mathbf{q}_{m=i}^t \quad (7)$$

where $\mathbf{q}_m^t$ indicates the runoff at time t for the MERIT unit basin m , as generated by the HBV model. All simulated runoffs from the upstream MERIT units of each training gage are summed into the main river runoff, $\mathbf{q}_b^{t'}$ (subscript b denotes the large gage basin), and then routed using the unit hydrograph internal routing formulas (Equations 8 and 9).

$$\mathbf{q}_b^{t'} = \int_0^t \xi(s) \mathbf{q}_b^{t-s} ds \quad (8)$$

$$\xi(s) = \frac{1}{\Gamma(\theta_{ra})\theta_{rb}^{\theta_{ra}-1}} s^{\theta_{ra}-1} e^{-\frac{s}{\theta_{rb}}} \quad (9)$$

where $\xi(s)$ is the gamma distribution-based unit hydrograph used for routing and Γ is the gamma function. q'_b is the simulated streamflow at the catchment outlet after channel routing; θ_{ra} and θ_{rb} are two routing parameters describing the shape of the hydrograph.

Additionally, HBV also can output soil moisture, SWE, and different components of streamflow. Similar to Song, Sawadekar et al. (2024), Kratzert et al. (2021), and Wilbrand et al. (2023), we utilized a normalized squared-error loss function (Equation 10) to enhance the high flow performance, compared to the weighted RMSE loss function used in the previous versions. Additionally, a capillary flux, as introduced by Song, Knoben et al. (2024), was incorporated to improve the low flow performance.

$$\text{Loss} = \frac{1}{B} \sum_{b=1}^B \frac{\sum_{t=1}^T (q'_b - q_b^*)^2}{(\sigma(q_b^*) + \epsilon)^2} \quad (10)$$

B is the total number of gage basins, T is the total of simulation days, q_b and q_b^* are respectively the simulated and observed runoff at the GAGES-II stations, $\sigma(q_b^*)$ is the standard deviation of the observed runoff in basin b (which avoids overweighting large and/or wet basins in the loss function), and ϵ is a small value to avoid a zero denominator. In contrast, $\delta\text{HBV1.0}$, as developed in Feng et al. (2022), assigned equal weight to the root-mean-square error (RMSE) of all basins in the loss function.

2.2.3. A Multiscale DM With Muskingum-Cunge Routing ($\delta\text{HBV2.0}\delta\text{MC}$)

Since the simple unit hydrograph routing formula overlooks the flow routing process and channel characteristics (essential elements for routing in large and complex river systems), an external differentiable Muskingum-Cunge routing model is incorporated to route flow across the high-resolution, continental, MERIT-based river network (derived from the ~ 90 m MERIT digital elevation model and flow directions; the median length of a river reach is ~ 7 km) and enhance streamflow prediction accuracy in large rivers. This model is called $\delta\text{HBV2.0}\delta\text{MC}$ -CONUS-hydroDL ($\delta\text{HBV2.0}\delta\text{MC}$ for short).

Both the purely data-driven LSTM model for streamflow and $\delta\text{HBV1.0}$ are lumped and need watershed boundaries and forcing/attribute inputs created for the whole upstream catchment of prediction points (gages), which is treated as one large basin. $\delta\text{HBV2.0UH}$ provides runoff (q_m) for each MERIT unit basin but still needs the full upstream catchment definitions to account for routing implicitly in order to obtain streamflow at specific points (gages). $\delta\text{HBV2.0}\delta\text{MC}$ provides discretized streamflow simulations at each reach in the river network. Unlike LSTM or $\delta\text{HBV1.0}$, it does not need the inputs for the whole upstream catchment—each MERIT incremental basin and reach uses its own inputs. Thus, we call $\delta\text{HBV2.0}\delta\text{MC}$ “seamlessly discretized” and use it to produce a data product for the entire CONUS river network (Figure 1c).

The differentiable version of Muskingum-Cunge (MC) routing utilizes embedded neural networks to learn hydraulic parameters as a function of river reach attributes (upstream catchment area, sinuosity, slope, etc.; each MERIT unit basin contains one reach) (Bindas et al., 2024). Our version is based on the δMC method developed by Bindas et al. (2024), with improvements to the efficiency of the routing algorithms. While earlier δMC methods solved Muskingum-Cunge explicitly on the CPU, we deployed an implicit scheme (David et al., 2011) running on the GPU to reduce computational time. Here, two Kolmogorov-Arnold Networks (KANs) (Z. Liu et al., 2024), KAN_h and KAN_l , were employed to estimate the hydraulic parameters and leakage parameters (Equations 11 and 12), respectively:

$$n_m, p_{c,m}, q_{c,m} = \text{KAN}_h(A_r) \quad (11)$$

$$\alpha, K/D, d_{gw} = \text{KAN}_l(A_l) \quad (12)$$

where n_m is Manning's roughness and $p_{c,m}$ and $q_{c,m}$ are channel geometric parameters of the MERIT unit basin, as described in Bindas et al. (2024). α is the leakance scaling coefficient, K is the saturated conductivity, D is the

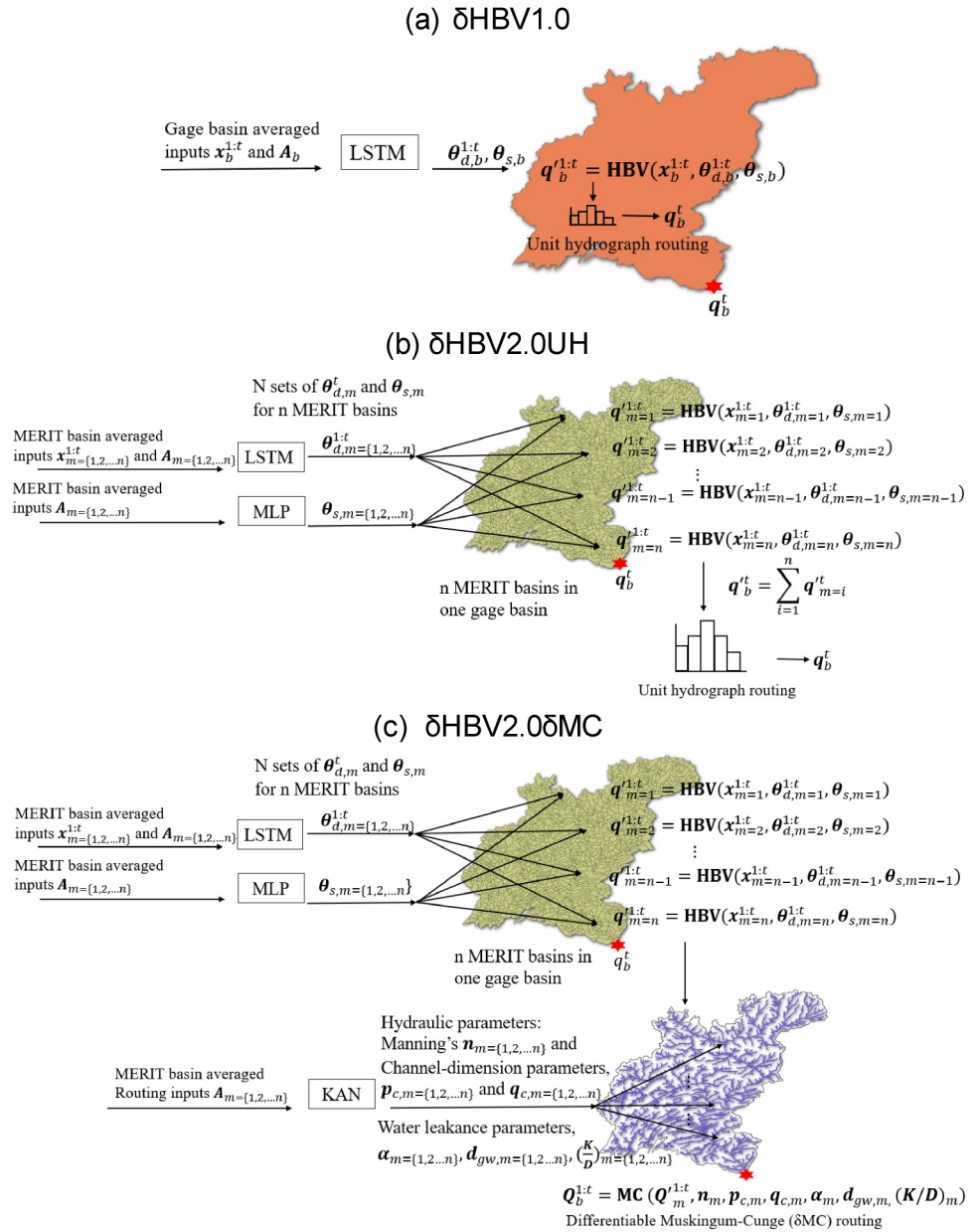


Figure 1. Schematic view of all differentiable models. (a) δ HBV1.0: coarse basin-averaged differentiable hydrologic model; (b) δ HBV2.0UH: high-resolution, multiscale differentiable hydrologic model using unit hydrograph routing for point or gage level prediction; and (c) δ HBV2.0 δ MC: high-resolution, multiscale differentiable hydrologic model using Muskingum-Cunge routing for seamlessly discretized streamflow simulation over 180 thousand MERIT unit basins.

diameter of the soil particles within the channel, and d_{gw} represents the depth of the groundwater table. $\mathbf{A}_r$ and $\mathbf{A}_l$ (components of which are listed as serving “channel geometry” and “leakage” modules, respectively, in Table A2 in Appendix A) are the static attributes used as inputs of KAN_h and KAN_l , respectively. KANs can approximate high-dimensional functions more efficiently than the MLPs used in Bindas et al. (2024), by combining univariate transformations (Z. Liu et al., 2024). The structures of the KANs are provided in Appendix B.

We forwarded the trained δ HBV2.0UH on all MERIT unit basins individually to obtain q_m as the forcing input of δ MC, along with parameters estimated by the KANs. For δ MC numerical stability, the q_m of each MERIT unit basin was mapped onto vectorized river flowlines with a uniform discretization of approximately 2 km. For each flowline, we solved a discretized MC equation with a finite difference scheme (Equation 13):

$$Q_{t+1} = c_1 I_{t+1} + c_2 I_t + c_3 Q_t + c_4 (Q - \zeta) \quad (13)$$

$$\zeta = \alpha \left(A_i \left(\frac{K}{D} \right) (d - d_{gw}) \right) \quad (14)$$

where I and Q are respectively the inflow and outflow of a flowline, each with the units of m^3/s . ζ is a water leakage parameter derived from the watershed storage state, similar to what was done previously by Shen et al. (2014). A_i is the incremental area of each flowline. Q is the lateral inflow of the incremental catchment area of the flowline, calculated with q_m^{t+1} from $\delta\text{HBV2.0UH}$. The flows at time step t are used as known variables to solve for the outflow at the next time step, Q_{t+1} . The Q_{t+1} for each flowline in the network is determined simultaneously by solving a matrix composed of one implementation of Equation 13 for each flowline. c_1 , c_2 , c_3 , and c_4 are the Muskingum-Cunge coefficients related to the hydraulic parameters from KAN_h , and differ from reach to reach and from time step to time step. More details can be found in Bindas et al. (2024).

2.3. Model Training and Evaluation

Both $\delta\text{HBV1.0}$ and $\delta\text{HBV2.0UH}$ were trained on 2,807 GAGES-II stations with relatively complete (>90%) daily observational time series data from 1 October 1980 to 30 September 1995. The gages of these basins are located within 500 m of the MERIT riverline. A similar basin group was used in Ouyang et al. (2021) and Song, Shen et al. (2024). $\delta\text{HBV2.0}\delta\text{MC}$ was trained from 1 October 1981 to 30 September 1995 on the same training basin group, while the first year (1980) of runoff was used as the warmup year for the simulations.

Both $\delta\text{HBV1.0}$ and $\delta\text{HBV2.0UH}$ were trained over 100 epochs with a mini-batch size of 100 and a time series length of 365. The LSTM in $\delta\text{HBV1.0}$ used a hidden size of 256 while the LSTM in $\delta\text{HBV2.0UH}$ used a hidden size of 64 since the latter is used only for dynamic parameterization for fewer HBV components. The MLP in $\delta\text{HBV2.0UH}$ used a hidden size of 4,096 for static parameterization, given its simple structure, more static parameters than dynamic ones, and a large number of basins. To favor simplicity and avoid overfitting, the hyperparameters were inherited from the previous $\delta\text{HBV1.0}$ model for a similar CONUS streamflow prediction scenario using the CAMELS data set or scaled based on the number of physical parameters output by the NNs (Feng et al., 2022; Song, Knoben, et al., 2024). We anticipate that some hyperparameter finetuning based on a well-selected validation data set can potentially result in moderately better models, but leave that to future work for a more comprehensive evaluation. δMC was only trained over five epochs with a mini-batch size of 64 basins, since the runoff generated by $\delta\text{HBV2.0UH}$ was already very close to the observations.

We evaluated all models on observations from 4,997 GAGES-II stations between 1 January 1980 and 31 December 2019, which were chosen according to the following criteria. There are 5,084 basins from GAGES-II where NWM3.0 streamflow simulation data is available for the CONUS. Of these, 4,997 have more than 10% valid daily observation data from 1 January 1980 to 31 December 2019. However, the observations included either in the training time period (1 October 1980 to 30 September 1995) of the 2,807 training basins used for the δHBV models or in the calibration time period (1 October 2016 to 30 September 2021) of the 1,365 CONUS calibration basins used for NWM3.0 were excluded from the evaluation set to avoid bias toward any of the models. The training and calibration gage locations are shown in Figure C1 in Appendix C. The NWM3.0 streamflow simulation data were aggregated from hourly to daily simulations before evaluation.

We tested the untrained variable, SWE, by comparing it with in-situ measurements from 525 SNOTEL sites from 1 January 2001 to 31 December 2019. The simulations of SWE from $\delta\text{HBV2.0UH}$ at the 525 SNOTEL sites were generated using inputs at the nearest grid points to SNOTEL sites from each input data set of the model and compared to SNOTEL observations (not directly using the SWE simulations of the nearest MERIT basins). This approach was only used for point prediction of SWE. The SWE predictions from the 1-km, 3-hr SWE product provided by NWM3.0 at the nearest grid points to the SNOTEL sites were aggregated to the daily scale and also compared to SNOTEL observations.

The following metrics were used for a comprehensive evaluation across all models: Spearman's correlation (called Spearman's rho), absolute relative total bias (ARTB), Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), normalized NSE (NNSE), root-mean-square error (RMSE) of peak flow (top 2% of flows),

Table 1
Statistics of Streamflow Metrics

Evaluation basins	Model	Median NSE	Median KGE	Median Spearman's rho	Median NNSE	Median low flow RMSE (mm/day)	Median peak flow RMSE (mm/day)
All 4,997 basins	NWM3.0	0.475	0.563	0.779	0.656	0.078	2.480
	δ HBV1.0	0.670	0.681	0.870	0.752	0.076	1.900
	δ HBV2.0UH	0.683	0.714	0.874	0.759	0.068	1.702
	δ HBV2.0 δ MC	0.670	0.710	0.857	0.752	0.062	1.790
	LSTM	0.679	0.720	0.896	0.757	0.054	1.848
1,836 “ ungauged ” basins	NWM3.0	0.382	0.467	0.741	0.618	0.073	2.195
	δ HBV2.0UH	0.519	0.555	0.818	0.675	0.062	2.006

Note. Bold values denote the best metrics in each column. The evaluation on all 4,997 basins excludes the training time period (from 1980-10-01 to 1995-09-30) of the 2,807 training basins used for δ HBV and the calibration time period (from 2016-10-01 to 2019-12-31) of the 1,635 CONUS calibration basins used for NWM3.0. The 1,836 “ ungauged ” basins are a subset of the 4,997 basins that were not used in either the training of δ HBV or the calibration of NWM3.0.

RMSE of low flow (bottom 30% of flows), peak bias, and correlation. The definitions of all metrics can be found in Table B2 of Appendix B.

3. Results and Discussion

3.1. CONUS-Wide Evaluation

Overall, the DMs supported by NN-based parameterization all improved upon NWM3.0 for the 4,997 GAGES-II basins according to various evaluation metrics, and they reached accuracy highly comparable to that of LSTM (Table 1). While the differences between the different δ HBV models, as well as LSTM, are barely noticeable, there is a clear gap between them and NWM3.0 on the NSE plot (Figure 2). Due to this similarity and the large overlap between δ HBV2.0UH and LSTM (δ HBV2.0UH almost completely overlaps with LSTM, which fills a gap seen in previous papers for poorly-performing basins (Feng et al., 2022; Song, Knoben, et al., 2024)), we will use δ HBV2.0UH as a representative in the remainder of this section. The noticeable performance jump to δ HBV2.0UH (median NSE = 0.683, Table 1) from NWM3.0 (median NSE = 0.475) despite the latter's extensive calibration effort is not simply due to the former having more training gages—its median NSE on 1,836 purely test (untrained) gages was 0.52 compared to NWM3.0's value of 0.38 on the same uncalibrated set (Table 1). Both models showed a decrease in performance when transitioning from the training data set to the testing data set, while δ HBV2.0UH demonstrated smaller differences and higher performance on the untrained time span and untrained gages (Table 1 and Table C1 in Appendix C). The NSE values of streamflow predictions for 3,990 out of the 4,997 basins were improved compared to those of NWM3.0 (Figure C2 in Appendix C). This large difference also cannot be explained by the different forcings used in these two models, as δ HBV2.0UH demonstrated similar performance under both forcing data sets on the training basins (Table C1 in Appendix C). We want to emphasize that NWM3.0 is a well-designed model with a more powerful base structure than HBV, and thus the true difference here is the NN-based parameterization and big data learning ability compared to traditional calibration. It can also be argued that being able to rapidly train on more gages is also an inherent advantage of DMs.

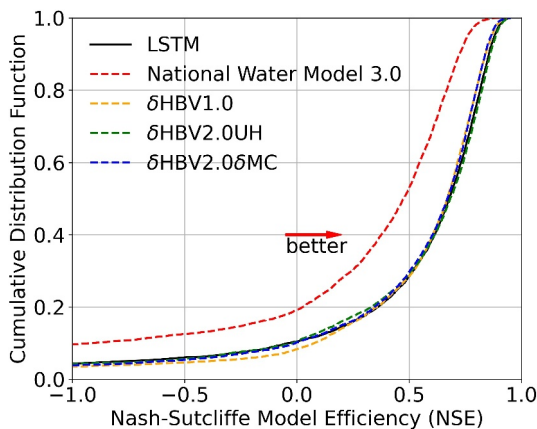


Figure 2. Cumulative distribution functions of Nash-Sutcliffe Efficiency for streamflow simulations from all models on 4,997 gage basins.

The improvements in DMs' metrics manifest in both flow bias and hydrograph shape according to the ARTB and Spearman's rho values, respectively (Figure 3). Spearman's rho can be used to detect monotonic trends in the shape of the time series. Tijerina-Kreuzer et al. (2021) considered a hydrograph with Spearman's rho greater than 0.5 as having “good shape.” The percentage of basins with “bad shape” was reduced from 14% with NWM3.0 to 9% with δ HBV2.0UH, and the percentage of basins with “large bias” (ARTB larger

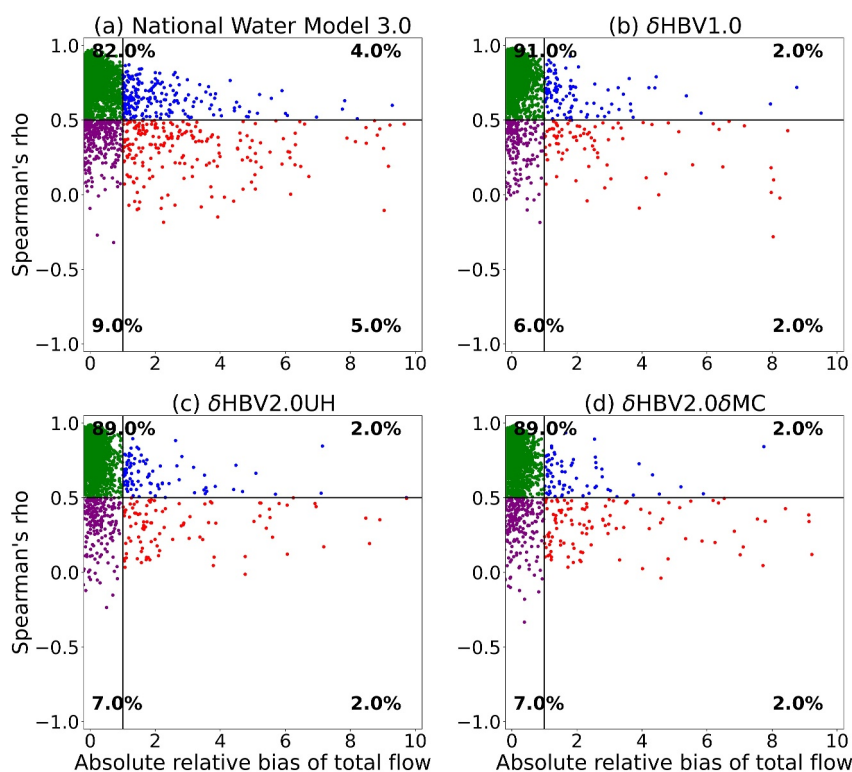


Figure 3. Spearman's correlation and absolute relative total bias (ARTB) plots comparing streamflow performance across 4,997 gage basins for (a) National Water Model 3.0, (b) δ HBV1.0, (c) δ HBV2.0 using internal unit hydrograph routing (UH), and (d) δ HBV2.0 routed by differentiable Muskingum-Cunge, δ MC. Horizontal and vertical lines show thresholds for Spearman's rho and ARTB, and each dot on a plot represents one basin. The green dots in the upper left corner denote the basins with high Spearman's rho and low ARTB ("good" category), while the red dots in the lower right corner denote the basins with low Spearman's rho and high ARTB ("bad" category). The bold text indicates what percentage of the 4,997 basins fall into each quadrant. The style of this plot was chosen to be comparable to Tijerina-Kreuzer et al. (2021).

than 1) was reduced by more than half, from 9% (Figure 3a) to 4% (Figure 3c). The green dots in Figure 3 represent basins with "good shape" (Spearman's rho) and "low bias" between modeled and observed streamflow. Approximately 90% of basins simulated by δ HBV models were in this category, compared to 82% for NWM3.0. To put things into context, Tijerina-Kreuzer et al. (2021) provided a similar evaluation for NWM 1.2 (but only on 2,200 basins), finding that 65% of them were in this category and 9% basins had low correlation and high bias. The improvements from NWM1.2 to NWM3.0 were due to monumental efforts in calibration (Cosgrove et al., 2024). The other DMs (δ HBV1.0 and δ HBV2.0 δ MC) produced comparable results to δ HBV2.0UH (Figures 3b and 3d).

To compare with the literature, we plotted NNSE (Cosgrove et al., 2024; Johnson et al., 2023; Nossent & Bauwens, 2012), which similarly shows a pronounced performance gap between NWM3.0 and the δ HBV models (Figure 4). δ HBVs had over 30% of basins with an NNSE value greater than 0.8 (compared to 6% with NWM3.0) and approximately 80% of basins had an NNSE over 0.6 (compared to 64% for NWM3.0). These results all indicate a major elevation in prediction accuracy due to NN-supported parameterization.

3.2. Regional Patterns and Comparisons Between DM Versions

The advantages of δ HBV2.0 models over NWM3.0 manifested as widespread positive changes in NSE values across the entire CONUS, including the eastern US and, more noticeably, the western US and the Prairie Potholes Region (PPR) in the northern central US (Figure 5). Among these regions, eastern basins were only moderately elevated by δ HBV2.0UH since NWM3.0 already simulated them well, while we found the largest NSE changes with large-area basins in the West and gages in the PPR. NWM3.0 shows some advantages in some poor-performing basins along the downstream Mississippi River and the southern Great Plains (Texas) (Figure 5d). These regions have high concentrations of reservoirs and human water use which exert major impacts on

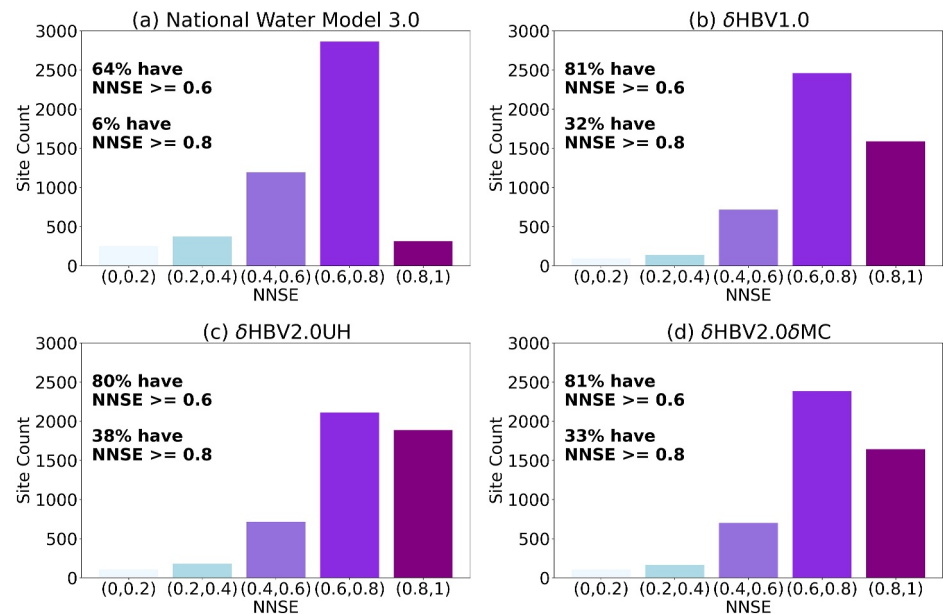


Figure 4. Normalized Nash-Sutcliffe Efficiency (NNSE) values for streamflow simulation across 4,997 gage basins for (a) National Water Model v3.0, (b) δ HBV1.0, (c) δ HBV2.0 using internal unit hydrograph routing (UH), and (d) δ HBV2.0 routed by differentiable Muskingum-Cunge, δ MC. The bold text indicates what percentage of the basins have NNSE values above a certain threshold. The style of this plot was chosen to be comparable to Cosgrove et al. (2024).

streamflows (Lehner et al., 2011). Such influences are simulated in NWM3.0 (Cosgrove et al., 2024) but not explicitly included in δ HBVs, which consequently could incur a noticeably high bias that could not be reconciled using either implicit or explicit routing schemes. All model performances were generally not satisfactory here in the southern Great Plains (many with NSE values below 0) and thus the metrics were less reliable (M. P. Clark et al., 2021), but δ HBV2.0UH seemed to perform worse than other models. It is worth noting that in this region, NWM3.0 benefitted from more calibration basins whereas δ HBVs had fewer training basins (Figure C1).

Compared to δ HBV1.0, the high-resolution, multiscale parameterization in δ HBV2.0 improved predictions in the northern and eastern US, which already had relatively high NSEs (Figure 5e). As a result, the higher-NSE part of the cumulative distribution function was improved for δ HBV2.0UH compared to δ HBV1.0, but the lower half was worse (Figure 2). The moderate improvement in the north likely reflected the benefit of better resolving the spatial distribution of snowfall and snowmelt, while the degradation in the south may be attributed to the high-resolution model estimating more runoff due to resolving spatial heterogeneity, which amplifies the peak flow bias due to the lack of water use modules. The PPR (Figure 5) has long been known as a challenge to all kinds of hydrologic models, whether process-based or ML-based (Feng et al., 2022; Kratzert et al., 2019; Song, Knoben, et al., 2024). In fact, the results from δ HBV2.0UH represent a rare achievement in median NSE for many of the basins here. δ HBV2.0 well captured the flow peaks at sites in this region, which were significantly underestimated by the basin-averaged δ HBV1.0 (Figure 6ii). We have two hypotheses for this improvement: (i) previous deficiencies were at least partially due to basin-averaged inputs not resolving the systematic spatial concentration of rainfall in parts of the domain, and/or (ii) the dynamic fast-flow recession coefficient (Section 2.2.2 and Appendix B) mimicks how wetland connectivity varies in time depending on climate conditions (Vanderhoof et al., 2016). The latter effect can also be described using formulations with more complex wetland storage dynamics (M. P. Clark & Shook, 2022), and future work with a more regional focus should examine the relative importance of these two processes. Besides the PPR, the high-resolution inputs of δ HBV2.0 benefited the mountainous area in the northwestern US, presumably by more accurately accounting for spatial heterogeneity in snowfall and better predicting high baseflows during the autumn (Figure 6i).

Stratifying results by catchment and flow regime, we see that δ HBV2.0UH had a greater advantage against the basin-lumped δ HBV1.0 for larger basins (Figure 7) and for higher flow percentiles (Figure 8), as it generally produced higher peak flows. The better simulations at higher flow percentiles were perhaps because it better resolved the heterogeneity of saturation and runoff thresholds. However, we still observed some performance

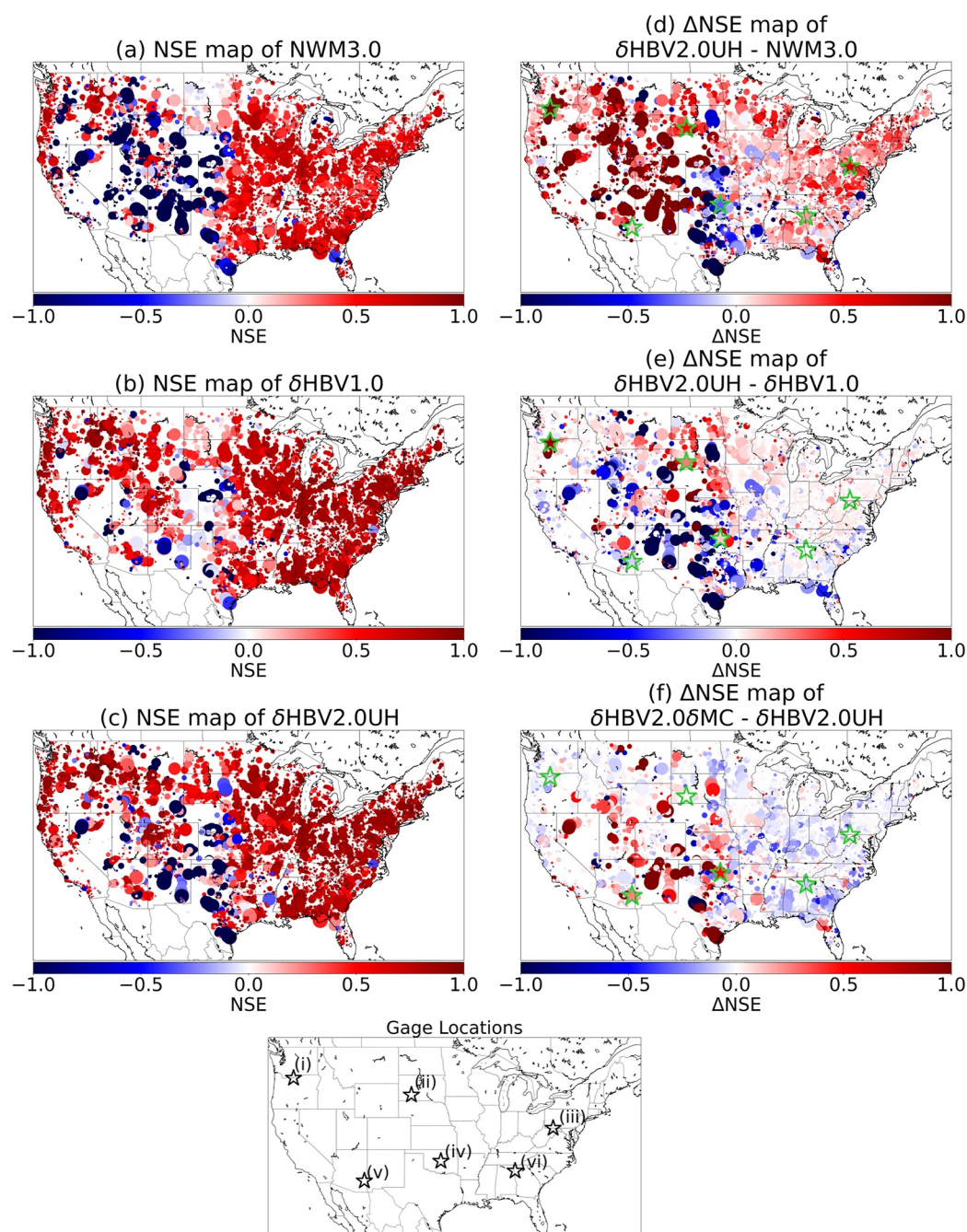


Figure 5. Nash-Sutcliffe efficiency (NSE) maps for (a) National Water Model v3.0, (b) δ HBV1.0, and (c) δ HBV2.0UH (δ HBV2.0 using internal unit hydrograph routing). NSE difference maps (d) between δ HBV2.0UH and National Water Model v3.0, (e) between δ HBV2.0UH and δ HBV1.0, and (f) between δ HBV2.06MC and δ HBV2.0UH. Red dots in panels (d–f) denote basins where the change is positive. The size of each dot indicates the basin's size. The sites in subplots d–f marked with star-shaped points (and labeled from i to vi) represent the gage locations of the hydrograph plots in Figure 6.

decreases in the southern US (Figure 5e), particularly in the poorly-performing downstream regions of large rivers in arid and semi-arid areas with reservoirs, where the sampling uncertainty is also high—meaning the metrics are less reliable as a means of diagnosis for interpreting the model results due to fewer data points being available (M. P. Clark et al., 2021). In this region, errors can also be larger due to anthropogenic impacts or the difficulty to capture flash floods. However, the lumped δ HBV1.0 model, with a larger influence from the NN and thus more flexibility, can more easily compensate for errors caused by unrepresented processes. Even though δ HBV1.0 has

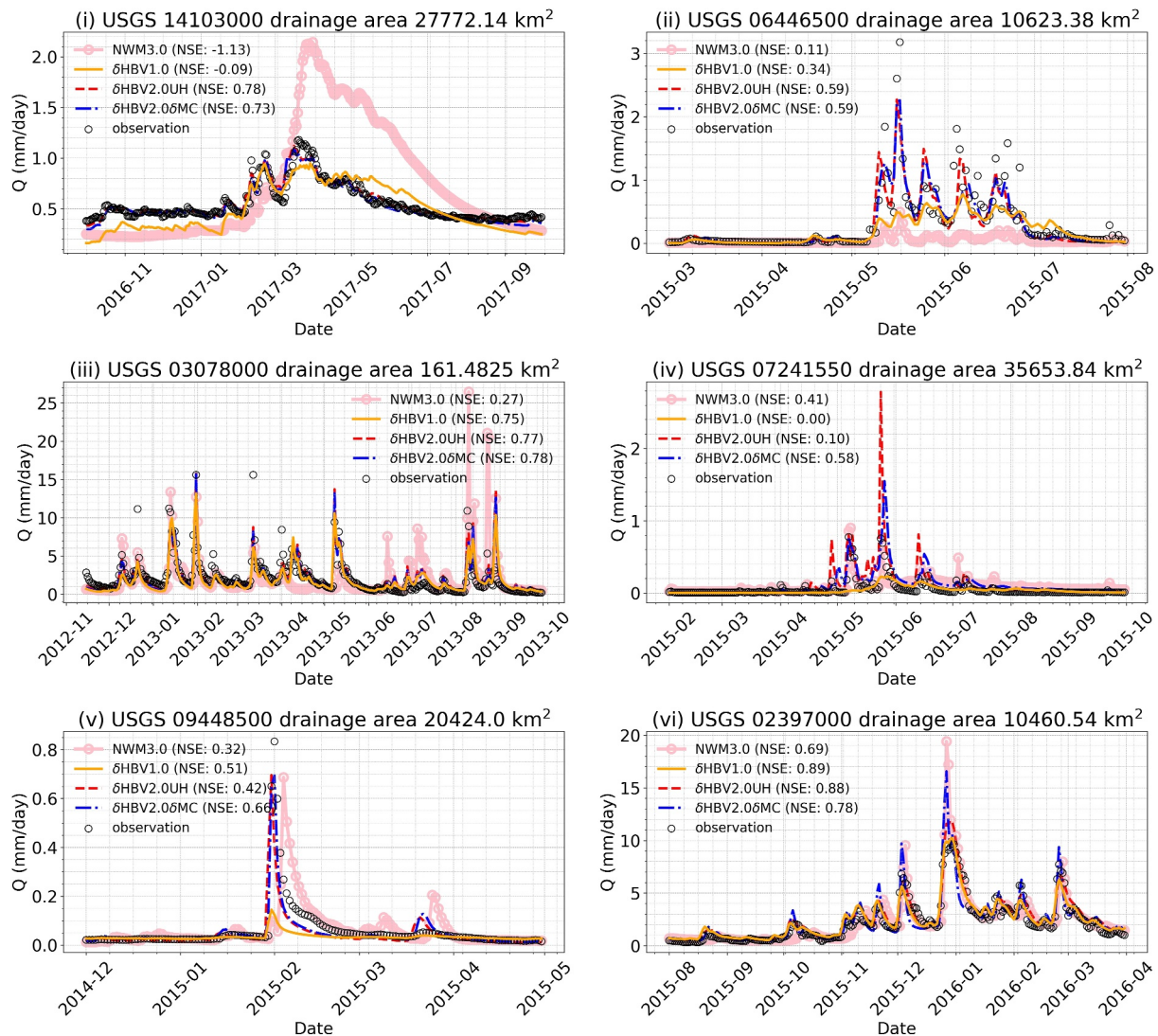


Figure 6. Observed and simulated hydrographs at locations annotated in Figure 5. (i) USGS 14103000: Deschutes River at Moody, near Biggs, OR; (ii) USGS 06446500: White River near Interior, SD; (iii) USGS 03078000: Casselman River at Grantsville, MD; (iv) USGS 07241550: North Canadian River near Harrah, OK; (v) USGS 09448500: Gila River at Head of Safford Valley, AZ; and (vi) USGS 02397000: Coosa River (Mayo's Bar) near Rome, GA. These sites were selected from the regions where (i, ii, and iii) δ HBV2.0UH, (iv) NWM3.0, or (iv and v) δ HBV2.06MC had advantages, or (vi) δ HBV2.06MC had disadvantages. Nash-Sutcliffe efficiency values of models in the whole testing period are listed in the subfigure legends.

some advantages in these regions, both models are still poor as discussed previously (Figures 5b and 5c). More importantly, we aim to provide a seamlessly discretized high-resolution streamflow product on the MERIT river network, which requires accurate runoff predictions at the fine-scale MERIT unit basins. However, δ HBV1.0, trained with aggregated inputs at the gage basin level, suffers from the well-known scaling issue (Gentine et al., 2012; Song, Shen, et al., 2024) and is not suitable for providing a seamlessly discretized product.

3.3. Impact of Explicit Routing

Even though the simulation performance of δ HBV2.06MC (with explicit Muskingum-Cunge routing) degraded slightly in humid regions such as the eastern US and the western coastal area relative to δ HBV2.0UH (Figure 5f), it showed an advantage in large basins in terms of KGE (Figure 7). This advantage existed particularly in the arid and semi-arid regions of the southwestern and central US (Figure 5f). Such an improvement was achieved by (i) better describing the flow conveyance for downstream river reaches, which allowed for accurate prediction of

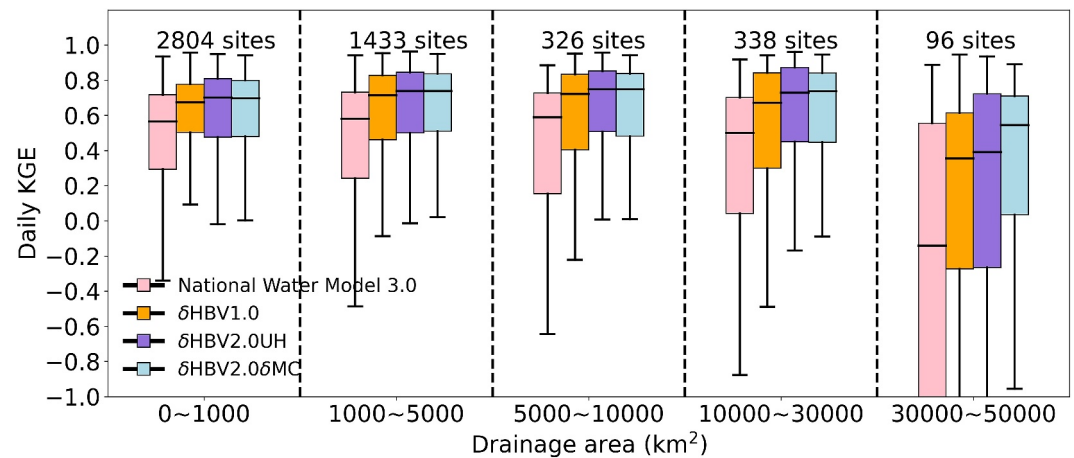


Figure 7. Daily Kling-Gupta Efficiency stratified by the drainage area of gage basins. Here, the horizontal line inside each colored box represents the median, while the top and bottom of the colored box indicate the first and third quartiles. The bars extending from the colored boxes indicate the lowest and highest data points within 1.5 times the interquartile range from the first and third quartiles.

peak timing, as observed in Figure 6v (one day difference with δHBV2.0UH). Additionally, (ii) the model accounts for water leakage as a loss from the river channel and the overall system. For example, the total amount of water was decreased by the routing model (Figure 6iv).

With δHBV2.0δMC, we can provide a seamlessly discretized CONUS-scale data set (Song & Shen, 2024) (Figure 9 shows the average flow in the 1995 water year) over the river network of ~180,000 MERIT unit basins from 1980 to 2020, without preparing basin-lumped forcings and attributes for every reach in the river network, as would be needed by δHBV1.0 or LSTM. This represents a top-tier streamflow data set with distinctly stronger metrics than models without NN-based multiscale parameterization (only slightly below δHBV2.0UH in terms of performance, as already discussed), including outputs of other hydrologic variables not used in training, such as evapotranspiration, baseflow, and SWE. When we train a basin-lumped model like δHBV1.0, the runoff has already incorporated scale-dependent regional-scale groundwater flow processes which can be prominent, especially in arid regions (Song, Shen, et al., 2024), and thus the output is only valid at basin outlets with the

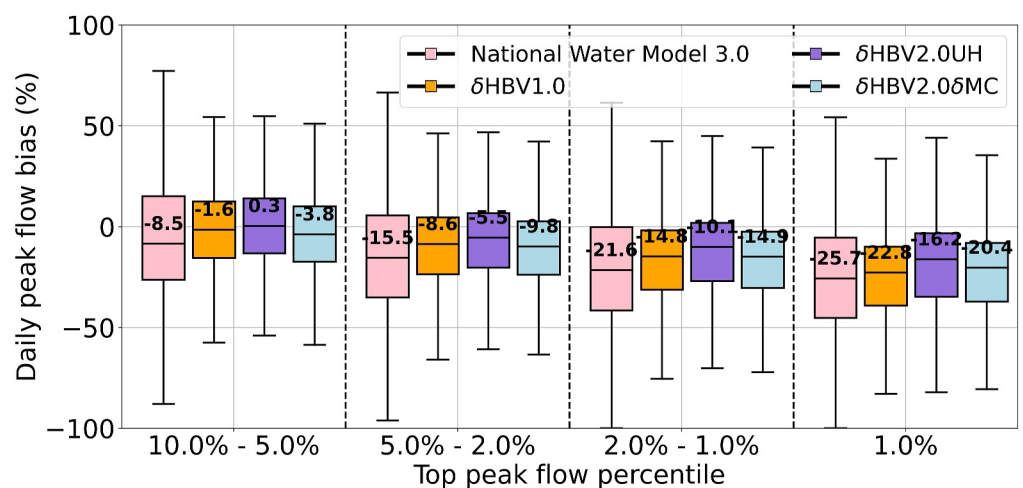


Figure 8. Mean daily peak flow bias percentage stratified by percentiles of the highest flows. For each basin, streamflow observations were sorted from high to low. The mean daily peak flow bias percentage was calculated for peak flows within specific percentile ranges. The mean values from all 4,997 gage basins were aggregated and presented in a single boxplot. Here, the horizontal line inside each colored box represents the median, while the top and bottom of the colored box indicate the first and third quartiles. The bars extending from the colored boxes indicate the lowest and highest data points within 1.5 times the interquartile range from the first and third quartiles.



Figure 9. The mean seamless daily streamflow simulation on the MERIT river network from 10/01/1995 to 09/30/1996, produced by δ HBV2.0 δ MC.

specified catchment area. On the other hand, when we seek to produce a high-resolution model, the model parameterization must be trained at that scale. Because we do not have many training stations that are at the MERIT basin scale at different parts of the CONUS, we are relying on the multiscale formulation and downstream gages to constrain MERIT-level parameters (Rakovec et al., 2019) with δ HBV2.0. The multiscale scheme does demand more computational resources than the basin-lumped δ HBV1.0 or LSTM, although it is still quite efficient compared with the cost of calibrating hundreds of basins individually with traditional methods. δ HBV2.0UH requires approximately 125 GPU hr and ~ 50 Gigabytes of GPU memory to train on 2,800 basins and 15 years of training data using an NVIDIA A100 GPU in a server hosted by the University of Alabama, while δ HBV1.0 needs 70 GPU hr and ~ 3 Gigabytes of GPU memory for the same training task. It only takes 1 GPU hr to forward δ HBV2.0UH across all $\sim 180,000$ MERIT unit basins for a 40-year simulation.

As side evidence of model internal dynamics, the point prediction of SWE simulated using model inputs from the nearest gridded points to the 525 SNOTEL stations demonstrated decent performance (daily NSE of 0.67, Table 2) when compared to SWE SNOTEL observations (Serreze et al., 1999). NWM3.0 provides a 1-km gridded SWE product from its land surface model, NoahMP, which reached an NSE of 0.52 in the same evaluation. Noticeably, the annual peak bias of SWE from δ HBV2.0UH was reduced to -9.6% , compared to -33.0% from NWM3.0, a difference which could matter greatly when forecasting spring water availability. Following previous studies (Broxton et al., 2016), an expert-prescribed constraint was adopted in δ HBV2.0 to differentiate temperature thresholds for rain and snow in MERIT unit basins in mountainous and plains regions, setting a constant 4°C threshold to increase snowfall above 2,000 m in elevation. Below this point, the temperature threshold is determined by the NN as used in δ HBV1.0 (Appendix B). The high-resolution δ HBV2.0 models can still benefit from additional human-prescribed constraints to accurately capture the heterogeneity of processes within large basins. It should be noted that neither NWM3.0 nor δ HBV2.0 were calibrated using SWE observations or satellite products, yet they still provide decent SWE predictions, although their performance is not as high as either process-based or data-driven models for SWE (Broxton et al., 2016; Hill et al., 2019; Song, Tsai, et al., 2024). In addition, δ HBV2.0 only accounted for basic snow accumulation and melting processes, without incorporating the effects of forest gaps, aerodynamic redistribution, changes in the energy balance due to impurities, etc. It was

Table 2

Statistics of Metrics for the Point Prediction of Daily Snow Water Equivalent at 525 SNOTEL Sites From 2001-01-01 to 2019-12-31

Model	Median daily NSE	Median daily correlation	Median annual relative peak bias (%)
NWM3.0	0.517	0.811	-33.0
δ HBV2.0UH	0.670	0.907	-9.6

trained using MERIT unit-basin-averaged inputs but tested with gridded inputs at specific sites to validate the simulation of internal states.

4. Further Discussion

This work advances hydrologic modeling via a multiscale differentiable hydrologic and routing model to produce operationally-ready streamflow predictions at a continental scale, while NWM3.0's strengths lie in the simulation of more physical processes and hydrodynamic equations with very high resolution. δ HBV2.0 models learned substantially from the NWM3.0 experiences on the impacts of high resolution. The differentiability of the δ HBV2.0 models enabled effective multiscale learning from large data to generate subbasin-scale parameters supervised by basin-scale observations. Even though the current DMs offer higher resolution compared to the previous versions, the rainfall-runoff processes are still simulated using conceptual models (like HBV). They focus on vertical infiltration processes (from the surface downwards into the soil), while lateral subsurface flow and baseflow movement are simplified as fast or slow water release from the soil, proportional to the water storage in the soil layers. In contrast, NWMs solve a two-dimensional (2D) St. Venant equation for overland flow (Julien et al., 1995) and a 2D saturated subsurface lateral transport equation for subsurface flow at a 250 m resolution (Cosgrove et al., 2024). Our current DMs mainly focus on capturing the overall behavior of the hydrologic system at a large scale and simplify flow mechanics by relying on aggregated physical parameters.

It is interesting that δ HBV1.0, using coarse basin-averaged forcings and attributes even for some 10,000–50,000 km² basins (Figure 7), already performed well compared to δ HBV2.0 without explicitly resolving basin heterogeneity. This is consistent with some earlier comparisons that were conducted at smaller scales (Smith et al., 2004). While δ HBV2.0UH and δ HBV2.0 δ MC led to improvements in large downstream gages and some regions like the PPR, the advances over δ HBV1.0 were not as ubiquitous as we had anticipated. The strength of basin-lumped δ HBV1.0 poses a challenge to hydrologic theories emphasizing spatial heterogeneity and nonlinear interactions, as well as arguments for hyperresolution simulation objectives at or below the 1 km scale (Wood et al., 2011). However, it is important to realize that this does not mean resolution does not matter—predictions are often required at finer resolution, and in fact our results show prominent impacts from scaling (Song, Shen, et al., 2024)—rather, it shows the NN had already adapted to the training data and compensated for the location-dependent scaling impacts. It also indicates there are probably generalizable scaling laws that can be leveraged.

Even though δ HBV2.0 δ MC still fell slightly behind δ HBV2.0UH (in other words, Muskingum-Cunge routing performed slightly worse than implicit unit hydrograph routing), a more complex external routing model is still necessary for modeling large rivers (Figures 6iv and 6v) and producing a seamlessly discretized product that covers all stream reaches. These modules should be viewed from an evolving perspective rather than a static one. The loss of accuracy could be due to numerical dispersion inherent with the Muskingum-Cunge formulation, some structural limitations in representing flood waves with the prismatic assumption, and static parameters, which can be addressed in the next development stage. A large data analysis calls for time-dependent Manning's roughness (Al Mehedi et al., 2024), which can be supported along with our other dynamic parameters. Furthermore, the unit hydrograph routing may be better at compensating for the impacts of reservoirs (Salwey et al., 2024) and human-related water uses. These modules should be explicitly added in the future, while addressing numerical issues and relaxing some fundamental limitations with the assumptions. Due to the spatial trade-off between different regions and basin scales in the large-scale training process, we observed that explicit routing benefited arid regions with large rivers in the southwest US, while harming other well-performing regions in the northern US (Figure 5f). A future mature parameterization scheme may relax these cross-region tradeoffs.

It is also currently not clear whether δ HBV2.0 accurately captures and simulates the heterogeneous hydrological systems within the basins when using high-resolution inputs supervised in an aggregate fashion at the catchment outlet. It is possible that the flow response at the subbasin scale is still underconstrained, and we invite the hydrological modeling community to evaluate this aspect. We hypothesize that a hybrid model with process-based elements can be better constrained than a fully data-driven multiscale model (J. Liu et al., 2022), but we leave a more detailed examination and potential improvements to future work. Additional model improvements could include the application of specific expert-prescribed constraints to regularize the model and improve the accuracy of untrained variables. Additionally, multivariate training or data assimilation could be used to reduce the distortion of fluxes within the MERIT unit basins.

In order to realize the goal of having both δ HBV1.0 and δ HBV2.0 be NWM-compatible, we have built two interfaces to facilitate the transition to potential deployment. One, a Python Model Interface (PMI), standardizes DM handling and development environments, including training and testing procedures. PMI attaches to a secondary interface, namely a Community Surface Dynamics Model System (CSDMS) Basic Model Interface (BMI) (Hutton et al., 2020; Peckham et al., 2013). BMI functions in accordance with the ongoing development of the Next Generation (NextGen) modeling framework, which will be the basis of future versions of the NWM. Critically, this BMI modularizes the δ HBV variants (with accommodation for future versions) so that they can be incorporated into, and operated within, the NextGen framework. While NextGen is still in development, having this interface architecture available enables δ HBV1.0 and δ HBV2.0 to be immediately compatible.

While the power of NN-derived parameterization has now been demonstrated, for fixed monitoring points with a long-term record, the traditional step-by-step calibration approach still has merit. Some of the US Regional Forecasting Centers (RFCs) still employ a manual, multi-step approach for a modest number of basins at scales over 300 km², and calibrate parameters affecting different parts of the hydrograph separately, for example, baseflow, SWE, and surface runoff. Future comparisons should be carried out with the RFC models.

5. Conclusions

In this study, we developed and evaluated coarse and multiscale differentiable hydrologic models (δ HBV1.0, δ HBV2.0UH, and δ HBV2.0 δ MC) over the CONUS. The DMs, supported by neural network-based parameterizations, leveraged high-resolution NWM modeling experiences and software pipeline. These models showed comparable skills in streamflow simulations to purely data-driven deep networks, with even better peak flow representations. They reached a median daily NSE of ~ 0.68 for 4,997 trained and untrained gage stations, nudging the majority of the basins' performances upward, and produced significantly smaller biases and better hydrograph shapes (according to Spearman's correlation). The multiscale parameterization (parameterized at MERIT unit basins but trained on USGS gages) enabled accurate runoff prediction at high resolution (MERIT unit basins, median 37 km²) to resolve the heterogeneity in forcing and basin attributes. The runoff was then routed downstream using either an implicit, unit-hydrograph approach (δ HBV2.0UH) or an explicit Muskingum-Cunge scheme (δ HBV2.0 δ MC). δ HBV2.0 δ MC, while currently experiencing a slight performance penalty compared to δ HBV2.0UH in smaller basins, can produce superior streamflow predictions in major downstream rivers and a seamlessly discretized hydrologic product.

We noted widespread performance improvements with δ HBV2.0 models compared to either NWM3.0 or δ HBV1.0 in both humid and arid regions, which were relatively slight in the already-high-performing northern CONUS, and more prominent in the western US and the Prairie Potholes Region, which are known for their modeling challenges. This could be due to improved representation of spatial concentrations in rainfall/snow, or the dynamical parameterization of fast-flow recessions which mimic the time-dependent connectivity of wetlands. The untrained SWE simulations were also highly reliable. The multiscale model still faces challenges with the southern Great Plains, which, as mentioned above, has extensive reservoirs and water uses. Future work should focus on the critical need of proper representation of reservoirs and water uses, further elevating ungauged basins, and addressing any numerical or algorithm issues with river routing schemes. It should also incorporate data assimilation approaches to improve short-term forecast capabilities.

Our models are designed to be compatible with the next generation of the NWM (NextGen) through the Basic Model Interface (BMI), ensuring seamless integration into operational water management frameworks. As such, they represent a promising direction for the future of continental-scale hydrologic modeling, balancing the need for high-resolution, accurate predictions with computational efficiency.

Appendix A: Data Details

The attributes variables used in the neural networks of δ HBV2.0 and δ MC are listed in Table A1 and Table A2, respectively.

Table A1
Attributes of Differentiable Models

Variables	Description [units]
catchment area/upstream area	Upstream area of a MERIT unit basin is used in δHBV2.0 and the gage basin area is used in δHBV1.0 [km ²]
ETPOT_Hargr	Potential evapotranspiration obtained using the Hargreaves method [mm/year]
FW	Fraction of open water data in the Global Lakes and Wetlands Database (Global Lakes and Wetlands Database, 2004) [-]
HWSD_clay	Proportion of clay in soil in the Harmonized World Soil Database (HWSD 1.2 (FAO et al., 2012) and 2.0 (FAO et al., 2023)) [%]
HWSD_gravel	Proportion of gravel in soil in the Harmonized World Soil Database (HWSD 1.2 and 2.0) [%]
HWSD_sand	Proportion of sand in soil in the Harmonized World Soil Database (HWSD 1.2 and 2.0) [%]
HWSD_silt	Proportion of silt in soil in the Harmonized World Soil Database (HWSD 1.2 and 2.0) [%]
NDVI	The Normalized Difference Vegetation Index (Dierckx et al., 2014) [-]
Porosity	Porosity from the Global Hydrogeology Maps (GLHYMPS) [-]
SoilGrids_clay	Proportion of clay in the SoilGrids data set (Poggio et al., 2021) [%]
SoilGrids_sand	Proportion of clay in the SoilGrids data set [%]
SoilGrids_silt	Proportion of clay in the SoilGrids data set [%]
Aridity	Aridity calculated from forcings
Glaciers	Glacier fraction from the Global Lithological Map (GLIMs, 2016) [-]
meanP	Mean precipitation from WorldClim (Fick & Hijmans, 2022) [mm/year]
meanTa	Mean temperature from WorldClim [°C]
Mean elevation	Mean elevation (Carabajal & Harding, 2006) [m]
meanslope	Mean slope (USGS et al., 2018) [-]
Permafrost	Permafrost [-]
Permeability	Permeability from the Global Hydrogeology Maps (Gleeson et al., 2014) (GLHYMPS) [log(m ²)]
seasonality_P	Precipitation seasonality index calculated using the Seasonality Index (SI) method (Walsh & Lawler, 1981) [-]
seasonality_PET	Potential Evapotranspiration seasonality index calculated using the Seasonality Index (SI) method (Walsh & Lawler, 1981)
snow_fraction	Snow cover fraction (Hall et al., 2006) [-]
snowfall_fraction	Snowfall fraction (Hall et al., 2006) [-]

Table A2
Attributes Used for Muskingum-Cunge Routing (δMC)

Parameter	Unit	Served modules	Data set
River Length	m	Channel Geometry	MERIT (Lin et al., 2021)
Direct River Length	m	Channel Geometry	MERIT
Sinuosity	–	Channel Geometry	MERIT
Slope	m	Channel Geometry	MERIT
Stream Drop		Channel Geometry	MERIT
Log of Upstream Drainage Area	km ²	Channel Geometry, Leakance	MERIT
Aridity	–	Channel Geometry, Leakance	WorldClim (Fick & Hijmans, 2017)
Alpha parameter for Mualem Van Genuchten Equation (ALPHA)	l/cm	Leakance	HiHydroSoil v2.0 (Poggio et al., 2021)
Fraction Clay (0–5 m)	%	Leakance	SoilsGrid 2.0 (Poggio et al., 2021)
Fraction Sand (0–5 m)	%	Leakance	SoilsGrid 2.0
Fraction Silt (0–5 m)	%	Leakance	SoilsGrid 2.0
Water content between saturation point and field capacity (pF2) (SAT_FIELD)	–	Leakance	HiHydroSoil v2.0
Porosity	–	Leakance	Global Hydrogeology Maps (GLHYMPS)
Organic Matter Content (ORMC)	%	Leakance	HiHydroSoil v2.0
N parameter for Mualem Van Genuchten Equation (N)	–	Leakance	HiHydroSoil v2.0
mean_elevation	m	Leakance	Global Multi-resolution Terrain Elevation Data (Danielson & Gesch, 2011) (GMTED2010)
Saturated Hydraulic Conductivity (Ksat)	cm/d	Leakance	HiHydroSoil v2.0
Percent glacier	–	Leakance	GLIMS: Global Land Ice Measurements from Space (GLIMS, 2016)

Appendix B: Method Details

The LSTM in Equation 1 and Equation 4 can be written as the following:

$$\begin{aligned}
 \text{Input transformation :} \quad & \mathbf{x}^t = \text{ReLU}(\mathbf{W}_I \mathbf{I}^t + \mathbf{b}_I) \\
 \text{Input node :} \quad & \mathbf{g}^t = \tan h(\mathcal{D}(\mathbf{W}_{gx} \mathbf{x}^t) + \mathcal{D}(\mathbf{W}_{gh} \mathbf{h}^{t-1}) + \mathbf{b}_g) \\
 \text{Input gate :} \quad & \mathbf{i}^t = \sigma(\mathcal{D}(\mathbf{W}_{ix} \mathbf{x}^t) + \mathcal{D}(\mathbf{W}_{ih} \mathbf{h}^{t-1}) + \mathbf{b}_i) \\
 \text{Forget gate :} \quad & \mathbf{f}^t = \sigma(\mathcal{D}(\mathbf{W}_{fx} \mathbf{x}^t) + \mathcal{D}(\mathbf{W}_{fh} \mathbf{h}^{t-1}) + \mathbf{b}_f) \\
 \text{Output gate :} \quad & \mathbf{o}^t = \sigma(\mathcal{D}(\mathbf{W}_{ox} \mathbf{x}^t) + \mathcal{D}(\mathbf{W}_{oh} \mathbf{h}^{t-1}) + \mathbf{b}_o) \\
 \text{Cell state :} \quad & \mathbf{s}^t = \mathbf{g}^t \odot \mathbf{i}^t + \mathbf{s}^{t-1} \odot \mathbf{f}^t \\
 \text{Hidden state :} \quad & \mathbf{h}^t = \tan h(\mathbf{s}^t) \odot \mathbf{o}^t \\
 \text{Output :} \quad & \mathbf{y}^t = \sigma(\mathbf{W}_{hy} \mathbf{h}^t + \mathbf{b}_y)
 \end{aligned} \tag{B1}$$

where $\mathbf{I}$ is the concatenation of the normalized meteorological forcings and static attributes. $\mathbf{W}$ and $\mathbf{b}$ with different subscripts are respectively the weight and bias of each layer. $\mathcal{D}$ is the dropout operator, which randomly sets some of the connections to zero to prevent overfitting. ReLU , $\tan h$, and σ represent the Rectified Linear Unit, tangential, and sigmoid activation functions, respectively. It is trained using the AdamDelta optimizer, which can automatically adjust the learning rate.

The MLP in Equation 5 has the following structure:

Table B1

The Model Structure of HBV Modules

Modules	Equations	Inputs or fluxes	Parameters
Hargreaves equation for potential evapotranspiration	$E_p = 0.00094(T_{\min} + 17.8)\sqrt{\Delta T R_a}$ $R_a = \frac{118.08}{\pi} d_r [\omega_s \sin(\phi) \sin(\delta) + \cos(\phi) \cos(\delta) \sin(\omega_s)]$ $\omega_s = \arccos(-\tan(\phi) \tan(\delta))$ $d_r = 1 + 0.033 \cos\left(\frac{2\pi N_{\text{day}}}{365}\right)$ $\delta = 0.409 \sin\left(\frac{2\pi N_{\text{day}}}{365} - 1.39\right)$ $\Delta T = T_{\max} - T_{\min}$	ϕ : latitude of gage. $T_{\min}$: daily minimum temperature. $T_{\max}$: daily maximum temperature. N_{day} : number of days in the year. R_a : solar radiation	No trained parameters
Snow accumulation and melt	$\frac{dS_p}{dt} = P_s + R_{fz} - S_{\text{smelt}}$ $\theta_{TT} = \theta_{TT0}$ if elevation < 2000 m, otherwise 4.0 °C $P_s = P \text{ if } T < \theta_{TT}, \text{ otherwise } 0$ $R_{fz} = (\theta_{TT} - T)\theta_{DD}\theta_{fz}$ $S_{\text{smelt}} = (T - \theta_{TT})\theta_{DD}$ $\frac{dS_{\text{liq}}}{dt} = S_{\text{melt}} - R_{fz} - I_{\text{snow}}$ $I_{\text{snow}} = S_{\text{liq}} - \theta_{CWH}S_p$	P : daily precipitation T : daily mean temperature S_p : snow storage. P_s : the precipitation as snow. R_{fz} : the refreezing of liquid snow. S_{melt} : the snowmelt as water equivalent. S_{liq} : the liquid water content in the snowpack. I_{snow} : snowmelt infiltration to soil moisture.	θ_{TT0} (range: [−2.5–2.5]), θ_{DD} (range: [0–0.1]), and θ_{fz} (range: [0.5–10]) are threshold temperature for snowfall [°C], degree-day factor [mm° C ^{−1} day ^{−1}], and refreezing coefficient [−], respectively. θ_{CWH} (range: [0–0.2]) is the water holding capacity as a fraction of the current snowpack [−].
Soil moisture and evapotranspiration	$\frac{dS_s}{dt} = I_{\text{snow}} + P_r - P_{\text{eff}} - E_x - E_T$ $P_r = P \text{ if } T > \theta_{TT}, \text{ otherwise } 0$ $P_{\text{eff}} = \min\left(\left(\frac{S_s}{\theta_{FC}}\right)^\beta, 1\right)(P_r + I_{\text{snow}})$ $E_x = (S_s - \theta_{FC})/dt$ $E_T = \min\left(\left(\frac{S_s}{\theta_{FC}\theta_{LP}}\right)^\gamma, 1\right)E_p$	S_s : the storage in soil moisture. S_{LZ} : the current storage in the lower subsurface zone. P_r : the precipitation as rain. P_{eff} : the effective flow to the upper subsurface zone. E_x : the rainfall excess. E_T : the actual evapotranspiration.	θ_C (range: [0–1]) is a time parameter [day ^{−1}]; θ_{FC} (range: [50–1,000]) is the maximum soil moisture (field capacity) [mm]; θ_{LP} (range: [0.2–1]) is the vegetation wilting point [−]; β (range: [1–6]) is a parameter influencing the shape of the soil moisture function [−]; γ (range: [0.3–5]) is a parameter influencing the shape of the evapotranspiration function [−].
Runoff generation	$\frac{dS_{UZ}}{dt} = P_{\text{eff}} + E_x - \text{perc} - Q_0 - Q_1$ $\text{Perc} = \min(\theta_{\text{perc}}, S_{UZ}/dt)$ $Q_0 = \theta_{K_0}(S_{UZ} - \theta_{UZL})$ $Q_1 = \theta_{K_1}S_{UZ}$ $\frac{dS_{LZ}}{dt} = \text{Perc} - Q_2 + R_T$ $R_T = \max\left(\min\left(\frac{\theta_{RT}(A_c - \theta_A)}{1000}, 1\right), -1\right)$ if A_c is smaller than 2,500 km ² , otherwise decay to 0. $Q_2 = \theta_{K_0}S_{LZ}$ $q' = Q_0 + Q_1 + Q_2$ $q(t) = \int_0^t \xi(s) q'(t-s) ds$ $\xi(s) = \frac{1}{\Gamma(\theta_{ra})\theta_{rb}} s^{\theta_{ra}-1} e^{-\frac{s}{\theta_{rb}}}$	S_{UZ} : the storage in the upper subsurface zone. perc : the percolation to the lower subsurface zone. Q_0 , Q_1 , and Q_2 are the near surface flow, interflow, and baseflow, respectively. R_T : lateral flow. S_{LZ} : the storage in the lower subsurface zone. q' and q are the total runoff before and after routing.	θ_{perc} (range: [0–10]) is the percolation flow rate [mm/day]; θ_{K_0} (range: [0.05–0.9]), θ_{K_1} (range: [0.01–0.5]), and θ_{K_2} (range: [0.001–0.2]) are the recession coefficients [day ^{−1}]. θ_{RT} (range: [0,20]) is lateral flow coefficient [mm km ^{−2} day ^{−1}]. θ_A (range: [0,2500]) is the area threshold [km ²]. θ_{ra} (range: [0–2.9]) [−] and θ_{rb} (range: [0–6.5]) [−] are two routing parameters. θ_{UZL} (range: [0–100]) is the maximum storage of the upper soil layer [mm].

$$\begin{aligned}
 \text{Input layer : } & \mathbf{x}_1' = \mathcal{D}(\text{ReLU}(\mathbf{W}_1 \mathbf{I}' + \mathbf{b}_1)) \\
 \text{Hidden layers : } & \mathbf{x}_i' = \mathcal{D}(\text{ReLU}(\mathbf{W}_h \mathbf{x}_{i-1}' + \mathbf{b}_h)) \\
 \text{Hidden layer : } & \mathbf{y}' = \sigma(\text{ReLU}(\mathbf{W}_o \mathbf{x}_i' + \mathbf{b}_o))
 \end{aligned} \tag{B2}$$

where $\mathbf{I}$ contains the normalized static attributes. $\mathbf{W}$ and $\mathbf{b}$ with different subscripts are the weight and bias of each layer. There are 6 hidden layers, and $\mathbf{x}_{i-1}'$ in each hidden layer represents the outputs from the previous layer. LSTM and MLP are trained together in δ HBV2.0. All outputs (parameters) from the LSTM and MLP are in the range of 0–1 due to the use of the sigmoid activation function. These outputs are then rescaled to the real parameter values using the parameter ranges provided in Table B1.

Table B2
Metrics Used for Model Evaluation

Metric	Definition	Formulas
Spearman's correlation (Spearman's rho)	The rank correlation between observed and modeled data to quantify the correlation of nonlinear and non-normally distributed data	$\rho = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)}$ d_i is the difference between the ranks of Q_i^{sim} and Q_i^{obs} N is the total number of data points in the time series
Absolute relative total bias (ARTB)	The relative bias between the total observed flow and the total simulated flow	$\text{ARTB} = \left \frac{\sum_{i=1}^N Q_i^{\text{sim}} - \sum_{i=1}^N Q_i^{\text{obs}}}{\sum_{i=1}^N Q_i^{\text{obs}}} \right $
Nash-Sutcliffe Efficiency (NSE)	The ratio of the model error variance to the observed variance, where 1 indicates a perfect model and 0 indicates performance equivalent to predicting with the long-term mean	$\text{NSE} = 1 - \frac{\sum_{i=1}^N (Q_i^{\text{sim}} - \bar{Q})^2}{\sum_{i=1}^N (Q_i^{\text{obs}} - \bar{Q})^2}$ where $\bar{Q}$ is the mean of Q_i^{obs}
Kling-Gupta Efficiency (KGE)	A comprehensive assessment of Pearson's correlation (γ), variability (α), and bias (β)	$\text{KGE} = 1 - \sqrt{(\gamma - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2}$
Normalized NSE (NNSE)	A variant of NSE with values ranging from 0 to 1, added to be comparable to previous papers	$\text{NNSE} = \frac{1}{2 - \text{NSE}}$
Root-mean-square error (RMSE) of peak flow	RMSE of the top 2% of the streamflow range	$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (Q_i^{\text{sim}} - Q_i^{\text{obs}})^2}{n}}$
Root-mean-square error of low flow	RMSE of the bottom 30% of the streamflow range	$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (Q_i^{\text{sim}} - Q_i^{\text{obs}})^2}{n}}$
Relative peak bias	The percentage of the peak bias	$\text{rBias} = \frac{Q_i^{\text{sim}} - Q_i^{\text{obs}}}{Q_i^{\text{obs}}}$
Correlation (γ)	Pearson correlation	$\gamma = \frac{\sum_{i=1}^N Q_i^{\text{sim}} Q_i^{\text{obs}} - N \bar{Q}^{\text{sim}} \bar{Q}^{\text{obs}}}{\sqrt{\left(\sum_{i=1}^N Q_i^{\text{sim}^2} - N \bar{Q}^{\text{sim}^2} \right) \left(\sum_{i=1}^N Q_i^{\text{obs}^2} - N \bar{Q}^{\text{obs}^2} \right)}}$

HBV in Equations 2 and 6 includes modules for simulating snow accumulation and melt, soil moisture, evapotranspiration, and runoff generation (Aghakouchak & Habib, 2010; Beck et al., 2020; Seibert, 1997). All modules, fluxes, and parameters in HBV are listed in Table B1.

The KANs used in Equations 11 and 12 can be broken down into the following equations:

$$\begin{aligned} \text{Continuous function : } \phi_{ij}(x_i) &= \alpha_{ij} \text{SiLu}(x_i) + \beta_{ij} \sum_i c_i B_i(x_i) \\ \text{KAN Layer : } K_j &= \sum_{i=1}^n \phi_{ij}(x_i) \\ \text{Output layer } y &= \sigma(K) \end{aligned} \quad (\text{B3})$$

where x_i is the input of node i of the input layer; subscript j denotes the node j of the output layer; $\sum_i c_i B_i(x_i)$ is the weighted sum of basis functions of all input nodes; B represents the B-spline function; c_i is the weight assigned to node i ; β_{ij} is the scaling parameter for node i used to calculate output K_j of the node j in the KAN layer; and SiLu denotes the SiLu activation function where $\text{SiLu}(x) = x/(1 + e^{-x})$. c_i , α_{ij} and β_{ij} are trainable parameters. More details about KAN can be found in Z. Liu et al. (2024).

The metrics used to evaluate all models are listed in Table B2.

Appendix C: Additional Results

The training and calibration stations, as well as the untrained and uncalibrated stations, are shown in Figure C1.

Figure C2 shows the scatter plot of NSE from δ HBV2.0UH and NWM3.0. δ HBV2.0UH improved the NSE values in 3,990 out of 4,994 evaluation basins. Table C1 provides their performance during the training and untrained time periods, as well as with different forcing variables. Both models showed decreased performance from training to testing, but δ HBV2.0UH exhibited higher training performance and a smaller decrease. δ HBV2.0UH performed better when using Daymet forcing compared to AORC (the forcing used in NWM3.0), though the difference is not significant.

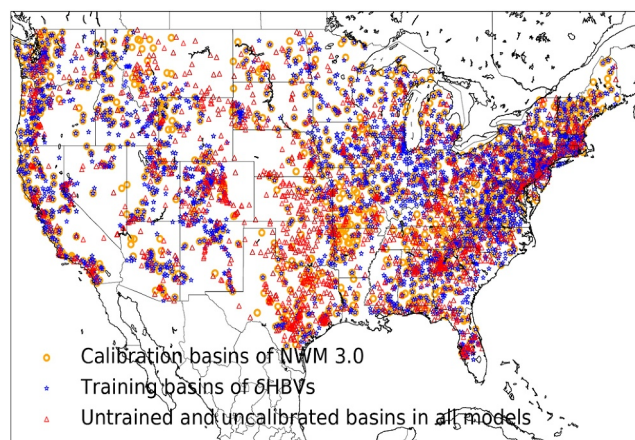


Figure C1. Map of calibration basins for NWM3.0 (orange rings), training basins for all δ HBV models (blue stars), and untrained and uncalibrated basins for all models (red triangles). NWM3.0 used 1,365 basins for calibration and all δ HBVs used 2,087 basins for training. There are 1,836 untrained and uncalibrated basins.

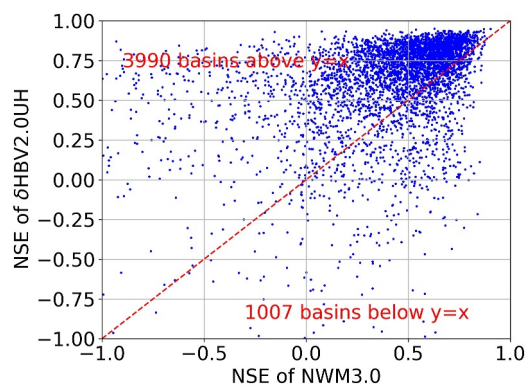


Figure C2. Scatter plot of Nash-Sutcliffe efficiency values from δ HBV2.0UH and NWM3.0 for 4,997 gage basins. Each blue point represents a single gage basin. The red dashed line represents a 1-1 relationship ($y = x$). There are 3,990 basins above this line, and 1,007 basins below this line.

Table C1
Streamflow Metrics of NWM3.0 and δ HBV2.0UH

Model	Evaluation basins	Evaluation time	Median NSE	Median KGE	Median Spearman's rho	Median NNSE	Median low flow RMSE (mm/day)	Median peak flow RMSE (mm/day)
NWM3.0	1,365 calibration basins	Calibration time span 2016/10/01–2019/12/31	0.692	0.804	0.829	0.764	0.073	2.270
NWM3.0	1,365 calibration basins	Testing time span 1980/01/01–2016/09/30	0.551	0.638	0.805	0.690	0.063	3.067
δ HBV2.0UH-Daymet	2,807 training basins	Training time span 1995/10/01–2019/12/31	0.814	0.850	0.897	0.843	0.056	1.12
δ HBV2.0UH-Daymet	2,807 training basins	Testing time span 1995/10/01–2019/12/31	0.763	0.787	0.899	0.809	0.068	1.381
δ HBV2.0UH-AORC	2,807 training basins	Testing time span 1995/10/01–2019/12/31	0.744	0.770	0.898	0.796	0.074	1.588

Note. Daymet is the forcing data used by δ HBV2.0UH while the Analysis of Record for Calibration (AORC) data set was developed by the National Weather Service Office of Water Prediction (Fall et al., 2023) and is used in NWM3.0.

Data Availability Statement

The streamflow data set (Song & Shen, 2024) from this work can be downloaded at <https://zenodo.org/doi/10.5281/zenodo.13774373> and is continuously augmented with more simulations. The multiscale DM, δ HBV2.0, which is implemented in a generic, scalable differentiable modeling framework, δ MG (Lonzarich et al., 2025), can be downloaded at <https://doi.org/10.5281/zenodo.14827983>. The δ HBV1.0 code is also available in δ MG. The SNOTEL data set (USDA NRCS NWCC, 2022) can be downloaded at <https://www.nrcs.usda.gov/wps/portal/wcc/home/>. Daymet forcing version 4 R1 (Thornton et al., 2022) is available at https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=2129. The fraction of open water data (Global Lakes and Wetlands Database, 2004) can be downloaded from www.worldwildlife.org/pages/global-lakes-and-wetlands-database. The Harmonized World Soil Database v1.2 and v2.0 (HWSD (FAO et al., 2012, 2023)) can be respectively downloaded from <https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v12/en/> and <https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v20/en/>. The Normalized Difference Vegetation Index (NDVI) data (Dierckx et al., 2014) can be downloaded from <http://www.vito-eodata.be/>. The GLObal HYdrogeology MaPS (GLHYMPS (Gleeson et al., 2014)) of permeability and porosity data can be downloaded from <https://borealisdata.ca/dataset.xhtml?persistentId=doi:10.5683/SP2/DLGXYO>. The SoilGrids and derived HiHydroSoil v2.0 data (Poggio et al., 2021) can be accessed from <https://gee-community-catalog.org/projects/isric/> and https://gee-community-catalog.org/projects/hihydro_soil/, respectively. The glacier fraction data can be extracted from the Global Lithological Map (GLIMS, 2016) <https://www.geo.uni-hamburg.de/en/geologie/forschung/aquatische-geochemie/glim.html>. The permafrost data (International Permafrost Association (IPA), 2010) can be obtained from <https://nsidc.org/data/g02190/versions/1>. The mean precipitation, temperature, and snowfall fraction can be obtained from the WorldClim data (Fick & Hijmans, 2017) <https://worldclim.org/>. The mean elevation (Carabajal & Harding, 2006) and slope data (USGS et al., 2018) can be respectively downloaded from <https://www.earthdata.nasa.gov/sensors/srtm> and <https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-global-30-arc-second-elevation-gtopo30>. The snow fraction data (Hall et al., 2006) can be downloaded from <https://nsidc.org/data/mod10cm/versions/5>.

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